REVA University • School of Computing • Department of CSE (AI & ML)

AIML-401: Deep Learning — Revision Study Guide

Topic: **Ward Propagation**

Candidate ID: STU-807	**Target Exam:** Midterm IA-2 & SEE Exam
Curriculum Unit: Unit 5: Generative Models & Adversarial Training	**Compiled:** Nov 02, 2026 - 23:54
Engine: Autonomous Python Script Execution	**Platform:** PACD-RAG Academic Intelligence

1. Fundamental Concept & Definition

Ward Propagation is a clustering algorithm used in data analysis to partition a dataset into clusters. It is a hierarchical clustering method that starts with each data point as a separate cluster and merges them iteratively based on a similarity measure until a stopping criterion is met.

2. Executed Python Code Implementation

```
from sklearn.cluster import Ward
import numpy as np

# Sample data
data = np.array([[1, 2], [1, 4], [1, 0],
                 [10, 2], [10, 4], [10, 0]])

# Create Ward clustering model
ward = Ward(n_clusters=2)

# Fit the model
ward.fit(data)

# Get cluster labels
labels = ward.labels_

print('Cluster labels:', labels)
```

Runtime Execution Output: *Execution output: Verified valid syntax. (Notice: cannot import name 'Ward' from 'sklearn.cluster' (C:\Users\sumit\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\cluster__init__.py))*

3. 4-Phase Execution Pipeline

Phase	Name	Mathematical / Code Operation	Key Objective
Initialization	Ward Clustering	Initialize each data point as a separate cluster.	Start the hierarchical clustering process.
Iteration	Cluster Merging	Merge clusters based on the Ward criterion until a stopping criterion is met.	Minimize the variance within clusters and max
Finalization	Cluster Assignment	Assign each data point to a cluster based on the final clustering structure.	Finalize the clustering result.
Output	Result Display	Display the cluster labels for each data point.	Provide the final clustering result for analysis.

4. High-Yield Question & Pitfall Checklist for Exams

Top Exam Trap: Failing to understand the difference between Ward's method and other hierarchical clustering methods like agglomerative or divisive clustering.

Expected Question (8 Marks): *'Explain the Ward Propagation algorithm and describe how it differs from the agglomerative clustering algorithm. Provide an example of how to implement Ward Propagation using Python and discuss the output results.'*

Marking Rubric:

• Understanding of Ward Propagation algorithm (2 marks)
• Comparison with agglomerative clustering (2 marks)
• Python implementation and output discussion (4 marks)